Hackathon Problem Statement 1: Unified Lifetime Medical History via Telemedicine in India (AI-Enabled)

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1. Introduction

India's healthcare ecosystem is inherently fragmented, with patients interacting with multiple providers across public hospitals, private clinics, pharmacies, and telemedicine platforms. Medical records are often distributed across paper prescriptions, siloed hospital systems, and informal digital channels such as messaging apps. This lack of continuity leads to incomplete patient histories, redundant diagnostics, inconsistent treatments, and increased risk of medical errors.

The rapid adoption of telemedicine in India presents a unique opportunity to address this fragmentation. Every teleconsultation generates digital data—text, voice, prescriptions—but much of it remains unstructured and underutilized. Without intelligent organization, these interactions fail to contribute to a persistent, longitudinal patient record.

This hackathon challenges participants to design and build an **AI-powered system that creates, maintains, and enhances a unified lifetime medical history for patients through telemedicine interactions**. The system should not only store records but also leverage AI to **structure, summarize, and derive insights** from each interaction, enabling better clinical understanding over time.

Participants are expected to simulate real-world telemedicine workflows and build a solution that evolves patient history across multiple consultations. With an extended development period, teams are encouraged to go beyond basic functionality and explore **multi-step AI pipelines, improved user experience, and meaningful clinical insights**, while still maintaining practical feasibility.

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2. Impact & Significance

Healthcare System Impact

• **Continuity across providers:** A unified longitudinal record ensures that patient history persists across different consultations and platforms.
• **Improved clinical efficiency:** AI-generated summaries and structured records reduce time spent on documentation and history review.

• **Early pattern detection:** Repeated symptoms, chronic conditions, or medication trends can be surfaced proactively.

Technology & AI Innovation

• **End-to-end AI-assisted workflows:** From raw input (text/voice) to structured medical records and insights.
• **Entity extraction and medical structuring:** Converting unstructured consultations into usable clinical data.
• **Context-aware summarization:** Generating concise yet meaningful patient histories across time.
• **Trend and risk signal identification:** Highlighting recurring issues or anomalies across consultations.

Societal Impact

• **Patient-centric healthcare:** Individuals gain access to clear, evolving medical histories.
• **Improved accessibility:** Telemedicine + AI lowers barriers in rural and underserved regions.
• **Foundation for national health systems:** Solutions can align with future integrations (e.g., ABDM ecosystem).

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3. Major Rules for Participation & Evaluation Criteria

A) Major Rules for Participation

1. **Expanded Scope (8-Day Build Expectation):**
   Participants are expected to build a **functional, end-to-end prototype** that includes:

- Patient profile creation and management
- Teleconsultation simulation (multiple visits)
- Persistent longitudinal medical history
- **Integrated AI pipeline (mandatory, more than basic usage)**
- Basic user interface for both doctor and patient views

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2. **AI Feature Requirement (Core Component):**
   Each solution must include **meaningful AI integration**, going beyond a single API call. Expected capabilities include one or more of the following:

- Automated summarization of consultation (text or voice)
- Extraction of key medical entities (symptoms, medications, diagnosis)
- Generation of a **cumulative patient history summary**
- Identification of **recurring conditions or trends across visits**
- Context-aware suggestions (e.g., highlighting past related cases)

Teams are encouraged to build a **simple AI pipeline** (input → processing → structured output → insight layer).

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3. **Recommended AI Setup (Important Note):**
   Participants are encouraged to use a **free LLM API such as Google Gemini (free API key)** for implementing AI features.

- Free-tier usage is sufficient for this challenge
- Other open/free models may also be used
- Focus should be on **effective application**, not model complexity

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4. **Multi-Consultation Requirement:**
   Teams must demonstrate at least **3–4 consultations per patient**, showing:

- Evolution of medical history
- AI-generated summaries improving over time
- Detection of patterns or continuity

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5. **Input Flexibility (Encouraged):**
   Participants may support one or more input formats:

- Text-based consultation notes
- Chat-style interaction
- Voice input (optional, bonus)

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6. **No Mandatory External Integrations:**
   Integration with real systems (ABHA, EHRs, insurance) is optional and may be mocked.

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7. **Data Privacy & Ethics:**

- Use only synthetic/dummy data
- AI outputs must avoid unsafe medical claims
- The system should act as **assistive**, not diagnostic

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8. **Submission Requirements:**
   Each team must submit:

- Working prototype (demo-ready)
- Description of AI pipeline and features
- Demo showing longitudinal history + AI outputs
- Optional: architecture or flow diagram

B) Evaluation Criteria

1) Functionality & Completeness (F)

- End-to-end working system
- Multiple consultations captured and linked
- Clear patient history creation

2) AI Depth & Effectiveness (A)

- Quality of AI outputs (summaries, extraction, insights)
- Use of AI beyond basic integration
- Ability to improve understanding of patient history

3) Longitudinal Intelligence (L)

- How well the system connects data across time
- Detection of patterns, trends, or recurring issues
- Quality of cumulative patient summary

4) Usability & Experience (U)

- Ease of use for doctors and patients
- Clarity of displayed information
- Integration of AI outputs into workflow

5) Practicality & Real-World Fit (P)

- Suitability for Indian healthcare conditions
- Simplicity, scalability, and adoption potential
- Consideration of constraints (connectivity, literacy)

6) Innovation & Design Thinking (I)

- Creative use of AI and system design
- Thoughtful features that go beyond baseline
- Vision for future expansion

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7) Demo & Communication (D)

• Clarity of demo
• Realistic patient journey
• Effective storytelling of impact

Final Evaluation

S = f(F, A, L, U, P, I, D)

Tie-breakers:

1. Stronger **longitudinal intelligence (pattern detection, continuity)**
2. More **practical and deployable solution**
3. Better **AI usefulness (not just complexity)**

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4. Success Definition

A successful solution is one where:

> A patient undergoes multiple teleconsultations, AI processes each interaction to extract and summarize key medical information, and the system builds a continuously evolving, intelligent medical history that provides meaningful insights and improves future care interactions.

Hackathon Problem Statement 2:
Innovating App-Based Solutions for Sustainable Agriculture

1. **Introduction:** Agriculture remains the backbone of the Indian economy, yet many farmers, especially in regions such as Eastern Uttar Pradesh, face persistent challenges including lack of timely crop advisory, limited market access, inefficient supply chains, and inadequate digital connectivity. Although several digital agriculture platforms exist, many solutions remain fragmented, providing either advisory services or market information but rarely integrating the entire farm-to-market ecosystem into a single accessible platform.

   To address these challenges, the App based initiative envisions a comprehensive mobile application-based AgriTech ecosystem that connects farmers, buyers, vendors, and government agencies while enabling data-driven agricultural decision-making.

2. **Challenge:** Design and develop an innovative mobile or web-based prototype inspired by the AgriTech concept that helps farmers improve productivity, decision-making, and market connectivity. The proposed solution should integrate smart farming, predictive analytics, and digital marketplaces into a single user-friendly platform.

3. **Expected Features:** Participants are encouraged to incorporate some or all the following features:

   a. Smart Farm Advisory System

      i. Provide farmers with crop advisory using weather data, soil conditions, and predictive analytics.

      ii. Include alerts for irrigation, pest risks, and crop health monitoring.

   b. Digital Farm-to-Market Platform

      i. Enable pre-harvest and post-harvest crop trading between farmers and buyers.

      ii. Include bidding or price discovery mechanisms to ensure fair market prices.

   c. AI/ML-Based Predictive Insights

      i. Predict crop prices, yields, or farm expenses using machine learning models.

      ii. Provide actionable insights to help farmers plan crop cultivation and sales.

   d. IoT and Sensor Integration (Optional)

      i. Demonstrate integration with IoT sensors (e.g., soil moisture, pH, NPK levels) for real-time farm monitoring.

   e. Localized and Inclusive User Interface

      i. Provide a simple interface with regional language support (e.g., Hindi or Bhojpuri) to improve usability for rural communities.

   f. Government and Vendor Services Integration

      i. Include features such as quality grading, input marketplaces, equipment rental, or government scheme awareness.

4. **Expected Outcome:** The hackathon aims to encourage participants to build scalable, farmer-centric digital solutions that leverage AI, IoT, data analytics, and mobile technologies to strengthen the agricultural value chain. The most impactful solutions will demonstrate how technology can create a smart, inclusive, and sustainable AgriTech ecosystem that empowers farmers and improves farm-to-market connectivity.

5. **Evaluation Criteria**

    a. Innovation and originality of the solution

    b. Practical applicability in rural agricultural environments

    c. Integration of AI, IoT, or data analytics

    d. User experience and accessibility for farmers

    e. Potential scalability and real-world impact

6. **Impact Vision:** The goal is to develop solutions that could evolve into platforms like AgriTech, enabling data-driven farming, transparent agricultural markets, and improved livelihoods for farmers across India.

Hackathon Problem Statement 3: AI for Missing Persons Identification & Human Trafficking Prevention in India

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1. Introduction

India faces a significant and persistent challenge of missing persons, with lakhs of cases reported every year, a large proportion of which involve children and vulnerable populations. Despite ongoing efforts by law enforcement agencies and non-governmental organizations (NGOs), many cases remain unresolved due to fragmented data systems, delayed identification processes, and lack of coordination across regions and platforms.

Information relevant to missing persons—such as CCTV footage, social media posts, public reports, and police records—exists in silos and is rarely analyzed collectively in real time. This gap creates opportunities for human trafficking networks to exploit delays and lack of visibility.

Advances in Artificial Intelligence, particularly in computer vision and natural language processing, present an opportunity to build systems that can **analyze diverse data sources, identify potential matches, and assist authorities in faster decision-making**.

This hackathon challenges participants to design and build an **AI-powered platform that helps identify, track, and assist in the recovery of missing persons by aggregating and analyzing multimodal data sources**. The system should simulate real-world workflows and provide actionable insights such as possible sightings, match probabilities, alerts, and location patterns.

Participants are encouraged to build a solution that is **practical, ethically responsible, and scalable**, while demonstrating meaningful use of AI across multiple data modalities.

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2. Impact & Significance

Societal Impact

• **Life-saving potential:** Faster identification and tracking can significantly increase the chances of rescuing missing individuals.
• **Prevention of human trafficking:** Early detection of movement patterns and suspicious activity can help disrupt trafficking networks.

• **Citizen participation:** Enabling public reporting can crowdsource valuable information in real time.

Law Enforcement & NGO Impact

• **Faster investigation cycles:** Automated matching and alerts reduce manual effort and response time.
• **Centralized intelligence:** Aggregating multiple data sources provides a unified view for decision-making.
• **Improved coordination:** Shared platforms can help police, NGOs, and authorities collaborate more effectively.

Technology & AI Innovation

• **Multimodal AI systems:** Combining computer vision, NLP, and geospatial analysis in a single workflow.
• **Real-world robustness:** Handling low-quality images, partial faces, and noisy text data.
• **Ethical AI design:** Balancing identification capabilities with privacy and responsible usage.

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3. Major Rules for Participation & Evaluation Criteria

A) Major Rules for Participation

1. **Expanded Scope (8-Day Build Expectation):**
   Participants are expected to build a **functional, end-to-end prototype platform** that includes:

- Missing person profile creation and management
- Data ingestion from at least one or more sources (images, text, or reports)
- AI-based matching, detection, or analysis
- Visualization dashboard and/or alert mechanism

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2. **AI Feature Requirement (Mandatory):**
   Each solution must include **meaningful AI integration**, such as:

- Face matching between missing person images and CCTV/public images
- Person detection in images/videos
- NLP-based extraction of potential sightings from text or reports
- Ranking or scoring of possible matches
- Linking multiple sightings across time or locations

Participants are encouraged to combine multiple AI techniques for stronger solutions.

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3. **Recommended AI Setup (Important Note):**
   Participants are encouraged to use a **free LLM API such as Google Gemini (free API key)** for tasks like:
- Text extraction and summarization
- Multilingual processing
- Report analysis

Other open-source models and tools are also allowed. Focus should be on **effective application rather than model complexity**.

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4. **Multi-Source Data Requirement:**
   Teams must demonstrate usage of **at least one data modality** (image or text).
   Use of multiple modalities (image + text + geospatial) will be considered an advanced implementation.

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5. **Data Simulation Requirement:**
   Participants may use publicly available datasets or synthetic data to simulate:
- Missing person profiles
- CCTV or image feeds
- Social media posts or public reports

Use of real sensitive or identifiable personal data is strictly prohibited.

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6. **Privacy & Ethical Constraints:**
   Solutions must demonstrate awareness of responsible AI usage:
- Avoid misuse of facial recognition
- Include safeguards where possible (e.g., access control, anonymization)
- Systems should assist authorities, not enable surveillance misuse

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7. **Submission Requirements:**
   Each team must submit:
- Working prototype (demo-ready)
- Description of AI components and approach
- Demo showing end-to-end workflow
- Explanation of ethical considerations

B) Evaluation Criteria

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1) Functionality & System Completeness (F)

• End-to-end working system
• Ability to manage missing person records
• Clear workflow from input to actionable output

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2) AI Effectiveness & Accuracy (A)

• Quality of detection, matching, or extraction
• Relevance and usefulness of outputs
• Handling of noisy or low-quality data

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3) Multimodal Intelligence (M)

• Integration of multiple data sources
• Ability to combine signals for better insights
• Logical linking of sightings or reports

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4) Practicality & Real-World Fit (P)

• Feasibility for Indian deployment conditions
• Usability for police/NGOs
• Consideration of infrastructure constraints

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5) Ethical AI & Responsibility (E)

• Awareness of privacy and misuse risks
• Inclusion of safeguards
• Responsible system design

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6) Innovation & Design Thinking (I)

• Creative problem-solving approaches
• Novel features (alerts, heatmaps, tracking graphs, etc.)
• Thoughtfulness in system design

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7) Demo & Communication (D)

• Clarity and flow of demonstration
• Realistic use-case simulation
• Ability to communicate impact effectively

Final Evaluation

S = f(F, A, M, P, E, I, D)

Tie-breakers:

1. Higher **real-world impact and usability**
2. Better **AI performance under challenging conditions**
3. Stronger **ethical and responsible design**

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4. Success Definition

A successful solution is one where:

> A missing person's profile is created, the system analyzes incoming data (images, text, or reports), identifies potential matches or sightings using AI, and presents actionable insights that can assist authorities in locating and rescuing the individual efficiently.

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5. References & Resources (For Participants)

To support development within the hackathon timeline, participants may use the following publicly available tools and datasets:

A) Government & Indian Data Sources

• Open Government Data Platform (https://data.gov.in)
• NCRB Crime Reports (https://ncrb.gov.in)
• NDAP (https://ndap.niti.gov.in)

B) Computer Vision & Face Recognition

• OpenCV – https://opencv.org
• face-recognition (Python) – https://github.com/ageitgey/face_recognition
• InsightFace – https://github.com/deepinsight/insightface
• DeepFace – https://github.com/serengil/deepface
• WIDER FACE Dataset – http://shuoyang1213.me/WIDERFACE

C) NLP & AI Tools

• Google Gemini API (Free Tier) – https://aistudio.google.com
• Hugging Face – https://huggingface.co
• spaCy – https://spacy.io

D) Geospatial & Visualization

• Leaflet.js / Mapbox / Google Maps APIs
• Kepler.gl – https://kepler.gl

E) Datasets for Simulation

• Kaggle (Missing Persons / Face Datasets) – https://kaggle.com
• CelebA / LFW datasets for face simulation
• Synthetic data generation using LLMs

Important Note

Due to privacy and security constraints, real datasets containing identifiable missing persons, CCTV feeds, or police intelligence systems are not publicly available. Participants are expected to use **synthetic or proxy datasets** to simulate real-world scenarios.

Problem Statement 4: Plant Disease Detection Using Leaf Images

Agriculture plays a critical role in global food security, yet plant diseases continue to significantly impact crop yield and quality. Participants are required to develop an AI-based system that detects plant diseases from leaf images while addressing real-world challenges.

Objectives

1. Classify plant leaf images into healthy and diseased classes, along with identifying reason.
2. Implement **at least** one innovation track (more is preferred always).

Innovation Tracks

Track A - Robustness: This track focuses on building models that perform reliably under real-world conditions rather than just on clean datasets. Participants should ensure their plant disease detection system can handle challenges such as noisy, blurred, or poorly lit images, as well as variations in backgrounds and environments. Techniques like data augmentation, training on diverse datasets, and testing under different conditions can help improve robustness, making the model more practical and dependable for real-world agricultural use.
Track B - Explainability: In this track, participants must go beyond predictions and make their models interpretable and trustworthy. The goal is to help users understand why a particular disease was predicted by highlighting important regions in the image or providing intuitive explanations, ensuring the AI system is transparent and acceptable to end users.
Track C - Lightweight AI: This track emphasizes designing models that are efficient and suitable for deployment on resource-constrained devices such as smartphones or edge devices. Participants should focus on reducing model size, improving inference speed, and minimizing computational requirements without significantly compromising accuracy. Approaches may include using efficient architectures like MobileNet, model pruning, or quantization, enabling real-time and accessible AI solutions in low-resource settings.
Track D - Synthetic Data Generation: This track encourages participants to address data scarcity by generating artificial training data to improve model performance. Since real-world datasets may be limited or imbalanced, participants can use techniques such as data augmentation or generative models to create diverse and realistic plant leaf images. The objective is to demonstrate that incorporating synthetic data can enhance model accuracy, generalization, and robustness, especially in scenarios with limited labelled data.

Dataset

Participants may use PlantVillage, PlantSeg or any other open datasets.

Deliverables

1. Code Repository
2. Report and Presentation Slides
3. Demo (Android app more recommended – See Plantix for example – also highlight the motivation, gaps and changes that you are bringing in your application.)

Evaluation Criteria

Innovation (50%), Demo (30%), Code Quality (20%)

Hackathon Problem Statement 5: Automated Annotation for Indic Manuscripts and Ram Charit Manas

1. Introduction

Indic manuscripts—on paper as well as palm-leaf—exhibit extreme variability in script and calligraphic styles, ink density, background texture, page geometry, and complex degradations such as bleed-through, stains, cracks, holes, fading, marginal notes, ornaments, and inconsistent ruling. Although large-scale digitization has produced extensive image collections, progress in Indic Document AI remains bottlenecked by the lack of high-quality annotations for core structures such as layout regions, text lines/baselines, damage masks, and reading order. Creating such ground truth manually is slow, costly, and expert-dependent, which prevents scaling datasets to the sizes needed for robust OCR/HTR, retrieval, and preservation pipelines.

Auto-Annotation proposes a challenge on automated annotation generation for Indian cultural heritage documents, combining diverse Indic manuscripts with a focused, culturally significant subset—Ramcharit Manas—to represent a realistic deployment scenario. The objective is to develop methods that produce usable pre-annotations directly from raw page images with minimal supervision, enabling humans to finalize annotations rapidly. Participants will be provided (i) a curated collection of manuscript pages across styles and degradation levels (including Ramcharit Manas pages), (ii) a small seed set of expert-annotated pages, and (iii) a larger unlabeled pool to encourage weak supervision, self-training, and active learning. Teams will submit predictions in a standard format (e.g., PAGE-XML or COCO polygons) for tasks including layout parsing (text regions, marginalia/notes, illustrations/diagrams, page/frame, and damage/holes) and text-line/baseline extraction. The key novelty of Auto-Annotation is its practical, human-centred evaluation. Beyond measuring annotation quality, we explicitly evaluate the human effort saved. On a hidden evaluation subset, submitted pre-annotations will be corrected by trained annotators using a fixed tool and protocol. On a hidden evaluation subset, submitted pre-annotations will be corrected by trained annotators using a fixed tool and

protocol. We will record correction time per page (seconds/page). Rankings will therefore reflect both accuracy and annotation efficiency, rewarding solutions that are not only correct but also fast to finalize. By benchmarking quality–cost trade-offs in a realistic setting, Auto-Annotation aims to catalyze scalable ground-truth generation pipelines, robust learning under severe manuscript degradations, and practical human-in-the-loop document understanding—accelerating the creation of high-value annotated datasets for Indic cultural heritage.

2. Impact & significance

Auto-Annotation addresses the primary bottleneck in Indic manuscript AI: scalable creation of reliable ground-truth annotations. While advances in layout analysis and HTR/OCR have improved model performance, real adoption on manuscripts remains limited due to the high cost, slow pace, and expertise required for annotation—especially under severe degradations and high stylistic diversity. By reframing the problem as automated annotation generation (rather than only recognition), and by including both a diverse Indic manuscript corpus and a focused, culturally significant subset (Ramcharit Manas), this challenge directly targets real-world dataset building and deployment needs.

Research Impact

- **A practical benchmark beyond accuracy:** Most document challenges evaluate only segmentation or OCR accuracy. Auto-Annotation introduces a Quality + Human-Effort evaluation, encouraging
methods that generate usable pre-annotations that are fast to finalize—an essential step for building large datasets.
- **Advances in low-label learning:** The setup of small seed labels + large unlabeled pool naturally promotes research in weak supervision, self-training, semi-supervised learning, uncertainty-aware predictions, and active learning—methods needed for heritage collections where labels are scarce.
- **Robustness under real degradations:** Manuscripts frequently exhibit holes/tears, bleed-through, ink spread, fading, warped pages, and non-uniform layouts. The benchmark explicitly drives solutions that generalize across materials (paper/palm-leaf), writing styles, and damage patterns—where standard document models often fail.
- **Stronger downstream pipelines:** High-quality auto-annotations unlock better training data for OCR/HTR, reading-order modeling, keyword spotting, retrieval, and translation workflows—improving end-to-end manuscript understanding and restoration pipelines.

Industry and Societal Impact

- **Direct value for libraries and archives:** Methods that reduce correction time can be integrated into digitization workflows, enabling institutions to convert large collections into structured, searchable corpora faster and at lower cost.
- **Cultural heritage preservation and access:** Improved annotations accelerate transcription and indexing, supporting scholars, educators, and the public in discovering and preserving Indic knowledge systems. The inclusion of Ramcharit

Manas strengthens cultural relevance and showcases a high-value, real deployment case.

• **Reusable community assets:** The challenge will release standardized formats (e.g., PAGEXML/COCO), evaluation scripts, and annotation guidelines, creating a long-lived benchmark and tooling ecosystem that can be adopted beyond the competition.

3. Major rules for participation & evaluation criteria

A) Major Rules for Participation

1. Data Policy (Core):
Participants may train using only the officially released Auto-Annotation dataset, consisting of a seed-labeled set and a larger unlabeled pool (covering both Indic manuscripts and Ramcharit Manas pages).

2. External Pretraining (Allowed with Full Disclosure):
Use of publicly available pretrained models/weights is allowed only if fully disclosed, including: model name, source link, and any external data used during pretraining. Use of private, proprietary, or undisclosed datasets is not permitted.

3. No Additional Manual Labeling:
Teams must not manually annotate any additional pages beyond the released seed labels. Any human labeling effort outside the provided seed set is prohibited to ensure fairness.

4. Submission Outputs & Format:
Submissions must follow the official schema and page identifiers, using PAGE-XML and/or COCOstyle JSON (final format confirmed at data release). Required outputs include:

- o Layout regions (instance polygons/masks) with class labels (e.g., text_region, marginalia/notes, illustration/diagram, page_frame, damage/hole).
- o Text lines as line polygons and optional baselines as polylines.

5. Submission Limits:
To prevent test-set overfitting, the number of test submissions will be limited (e.g., 1-3 submissions per team). Validation submissions will be allowed as per organizer policy.

6. Reproducibility (Required for Winner Verification):
Shortlisted teams must provide a brief method description and runnable inference instructions (script/container) to enable verification.

7. Fair Use & Ethics:
The dataset is provided strictly for research and challenge participation. Teams must respect dataset licensing, avoid attempts to re-identify sources beyond provided metadata, and ensure submissions/releases do not contain offensive, harmful, or sensitive content.

B) Evaluation Criteria

Auto-Annotation evaluates submissions on two complementary dimensions: annotation quality and human effort required to finalize annotations.

1) Annotation Quality Score (Q)

Computed on the hidden test set using official evaluation scripts.

Track 1: Layout Auto-Annotation

- Region segmentation quality: mIoU and/or mask mAP (as applicable)
- Instance-level quality: Panoptic Quality (PQ) where applicable

Track 2: Text-Line / Baseline Auto-Annotation

- Line extraction: Line IoU and/or Line F1-score
- Baseline accuracy: tolerance-based baseline distance score (if baselines are provided)

Note: The final class list, output schema, and official metric definitions will be released with the
dataset and evaluation toolkit.

2) Human Effort Score (E) — Time-only

Measured on a hidden Human-in-the-Loop subset. Trained annotators will correct each team's preannotations to match the official annotation guidelines using a fixed tool and protocol (same UI, same instructions). We record:

$$E = \text{Median(Time/page)}$$

where Time/page is measured in seconds per page. Median is used to reduce the impact of outliers (extremely hard pages).

Fairness controls: same annotator pool across teams, randomized page order, and a balanced mix of both domains (Indic manuscripts + Ramcharit Manas) and degradation levels.

3) Final Ranking

We will provide one of the following transparent approaches:

Pareto Ranking

Leaderboard highlights methods that dominate others in both higher quality (Q) and lower effort
(E).

Combined Score (Single Number)

$$S = Q \times \exp(-\lambda E)$$

(or an equivalent normalized weighted combination; λ published in advance).

Tie-breakers:

1. Higher performance on the Hard Subset (severe bleed-through, holes/tears, low contrast),
2. Lower human effort (E),
3. Faster inference runtime (optional, if standardized).

The Datasets for the challenge are
RamCharit-Manas:
https://dn721901.ca.archive.org/0/items/ramcharitmanas_202310/ramcharitmanas.pdf
https://archive.org/details/361952998SrimadValmikiRamayanaSanskrit1933AD/page/n19/mode/2up

Indic Manuscript Dataset: https://openn.library.upenn.edu/html/indic_contents.html